13.2.4 Digestive System

The digestive system includes the **alimentary canal** and the **associated digestive glands**. The alimentary canal consists of mouth, buccal cavity, pharynx, oesophagus, stomach, small intestine, caecum, large intestine and anus.

Mouth is a transverse slit bounded by upper and lower lips. It leads into the **buccal cavity**. The floor of the buccal cavity is occupied by a muscular tongue. Jaws bear teeth.

The buccal cavity leads into the oesophagus through the pharynx. **Oesophagus** opens into the stomach followed by small intestine. **Caecum** is a thin walled sac present at the junction of small intestine and large intestine. It contains bacteria that helps in **digestion of cellulose**. The **small intestine** opens into the **large intestine** which has **colon** and **rectum**. The rectum finally opens outside by the anus.

Digestive glands

The digestive glands are salivary glands, gastric glands, liver, pancreas and intestinal glands. The secretions of digestive glands help in digestion of food in the alimentary canal.

Dentition in Rabbit

Teeth are hard bone-like structures used to cut, tear and grind the food materials. The rabbit has two sets of teeth. The existence of two sets of teeth in the life of an animal is called **diphyodont dentition**. The two types of teeth are **milk teeth** (young ones) and **permanent teeth** (in adults).

In rabbit the teeth are of different types. Hence, the dentition is called **heterodont**. There are four kinds of teeth in mammals viz. the **incisors** (I), **canines** (C), **premolars** (PM) and **molars** (M). This is expressed in the form of a dental formula.

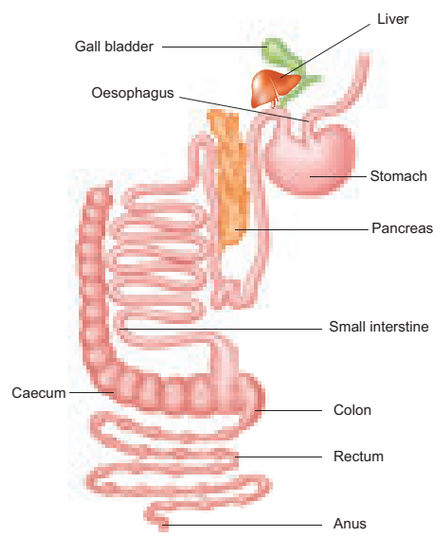


Figure 13.6 Digestive System of Rabbit

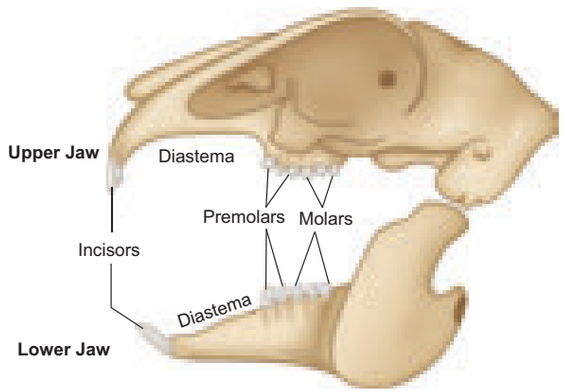


Figure 13.7 Dentition of Rabbit
(Arrangement of teeth in jaws)

Dental formula is the simple method of representing the teeth of a mammal. The number of each kind of tooth in the upper and the lower jaws on one side is counted.

Dental formula is ($I\ \frac{2}{1}$, $C\ \frac{0}{0}$, $PM.\ \frac{3}{2}$, $M\ \frac{3}{3}$.) in rabbit which is written as $\frac{2033}{1023}$. Canines are absent. The gap between the incisors and premolar is called **diastema**. It helps in mastication and chewing of food in herbivorous animals.

13.2.5 Respiratory System

Respiration takes place by a pair of **lungs**, which are light spongy tissues enclosed in the thoracic cavity. The thoracic cavity is bound dorsally by the vertebral column and ventrally by the sternum, laterally by the ribs. On the lower side of the thoracic cavity is the dome shaped **diaphragm**.

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